

Lab Exercise #5: Guessing Game

1 Purpose

This lab will give you practice using for loops, decision-making (if/else, switch), and logical operators to build an interactive program.

2 Overview

- Write a C program and have a user guess the number.
- The program will have a secret number between 0 and 50 (you may hard code the secret number).
- Ask the user to input the number of attempts (use the variable name `numAttempts`).
- Use a menu to allow the user to select play the game or quit.
 1. Play the game
 2. Quit
- After each guess:
 - If the guess is too high, print "Too high! Try again."
 - If the guess is too low, print "Too low! Try again."
 - If correct, print "Congratulations! You guessed it!" and stop the loop early with `break`.
- After the loop ends, if the user didn't guess the number, display "Sorry, you're out of attempts. The number was ...".

3 Code Documentation

The very first lines in *all* of your lab project files this semester should be comments in the class header file (copy from `header.c`). Of course, the items in the square brackets `[]` in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, for the lab exercises, should be the name of the assignment, your name and your lab partners name, e.g:

```
printf("Lab #5 - Joe Student and Jane Doe\n");
```

The next line of output should be `printf("EGRE 245: Introduction to Programming Using C, September 23\n");`

Please fill in the correct date.

4 Sample Run

Your output should match the sample run below.

```
Mark Schwitzerlett and Chilly Willy (the cartoon penguin)
Lab #5 -- September 30, 2025

Welcome to the Guessing Game!
1. Play Game
2. Quit

Enter your choice: 1

How many attempts would you like? 5

Guess the number (between 1 and 50): 22

Too high! Try again.

Guess the number (between 1 and 50): 10

Too low! Try again.

Guess the number (between 1 and 50): 15

Too low! Try again.

Guess the number (between 1 and 50): 18

Too high! Try again.

Guess the number (between 1 and 50): 17

Congratulations! You guessed it in 5 attempt(s)!

...Program finished with exit code 0
Press ENTER to exit console.
```

5 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Please name your file `Lab5_FirstName.LastName.c`.

Due date: The lab is due at the end of your lab period, either Tuesday, September 30th, or Thursday, October 2nd.